

BioMaster 首轮湿实验候选详细评审报告

逐项理由、可行性、风险与已有研究排重

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1 报告范围与结论

本报告面向进入首轮湿实验或最终人工复核的 12 个优先候选。候选来自当前 BioMaster 多证据排序与 final pre-experiment gate：亲和筛选、疾病证据、TxGNN/KG、Reactome/GO/CREEDS、CMap/LINCS 反转、GTEx/HPA/DepMap、结构对接、ADMET 与实验可执行性共同参与筛选。

需要强调：这里的结论是计算优先级与实验设计建议，不是生物学验证结论。DiffDock 数值用于姿态和口袋审计，不等同于结合自由能；PubMed/ClinicalTrials 自动检索用于提示排重，不能替代专家逐篇核查、专利检索和具体适应症审阅。

- 进入详细评审的候选：12 个；其中 {” 新用途或新靶点候选”：5, ” 机制延展”：4, ” 阳性对照/已知机制”：3}。
- 疾病方向覆盖：{” 肿瘤”：6, ” 神经/精神”：4, ” 心血管”：2}。
- 明确已知机制/阳性对照：3 个，主要用于实验体系校准和召回能力展示，不作为新发现候选宣传。
- 自动检索提示已有较多公开线索：Gefitinib-KIT，应标注为机制延展或二线验证；若专家更重视新颖性，可由 backup 候选替换。
- 新颖性较高且值得优先讨论的候选包括 Fludarabine-ADORA1、Palonosetron-F2R、Methylnaltrexone-OXTR/CRHR1、Donepezil-EDNRA、Amisulpride-ADRA1A 等，但这些候选必须通过正交靶点 engagement、counterscreen 与非毒性剂量窗口验证。

2 筛选与排重口径

筛选逻辑不是单一亲和排序，而是先在 druggable proteome 内完成药物-蛋白组合评分，再按疾病方向证据、机制路径、签名反转、组织/模型背景、结构可解释性、ADMET 风险和实验可执行性收敛。最终 gate 不直接决定购买，购买前仍需要确认化合物形态、纯度、供应商、溶解度、临床可达游离暴露、目标蛋白在实验模型中的表达、调控方向、正交实验和 counterscreen 可用性。

已有研究排重采用两层口径：第一层为本地 FDA/ChEMBL 已知 drug-target 标签；第二层为在线 PubMed 和 ClinicalTrials 自动查询。PubMed 的 drug-target 共现用于提示是否已有直接组合线索，ClinicalTrials 的药物-疾病方向计数仅说明该药在该领域是否常见，不等同于该靶点机制已经被验证。

3 候选总览

Rank 候选	方向	角色	分数	已有研究信号	处理建议
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1	Gefitinib - KIT	肿瘤	机制延展	96.7	已有较多公开共现线索	不建议作为纯新发现表述; 应标注为机制延展/二线验证。只有在实验能排除 EGFR carryover 并证明 KIT engagement 时才值得推进。
2	Fludarabine Phosphate - ADORA1	肿瘤	新用途或新靶点候选	94.8	未见明显直接组合文献	可作为新用途/新靶点候选推进; 购买前仍需 PubMed、ClinicalTrials 和专利/中文文献人工复核。
3	Gefitinib - EGFR	肿瘤	阳性对照/已知机制	93.5	已知机制/阳性对照	不作为新机制发现宣传; 建议保留为实验体系校准、阳性对照和召回能力证明。
4	Dacomitinib - EGFR	肿瘤	阳性对照/已知机制	93.2	已知机制/阳性对照	不作为新机制发现宣传; 建议保留为实验体系校准、阳性对照和召回能力证明。
5	Cladribine - ADORA1	肿瘤	机制延展	92.7	有少量文献线索	保留候选, 但购买前需逐篇确认是否已完成直接靶点验证或只是同领域背景共现。
6	Palonosetron Hydrochloride - F2R	肿瘤	新用途或新靶点候选	87.1	未见明显直接组合文献	可作为新用途/新靶点候选推进; 购买前仍需 PubMed、ClinicalTrials 和专利/中文文献人工复核。
7	Methylnaltrexone Bromide - OXTR	神经/精神	新用途或新靶点候选	85.8	未见明显直接组合文献	可作为新用途/新靶点候选推进; 购买前仍需 PubMed、ClinicalTrials 和专利/中文文献人工复核。
8	Methylnaltrexone Bromide - CRHR1	神经/精神	新用途或新靶点候选	85.0	未见明显直接组合文献	可作为新用途/新靶点候选推进; 购买前仍需 PubMed、ClinicalTrials 和专利/中文文献人工复核。
9	Donepezil Hydrochloride - EDNRA	Hy- 心血管	新用途或新靶点候选	84.7	未见明显直接组合文献	可作为新用途/新靶点候选推进; 购买前仍需 PubMed、ClinicalTrials 和专利/中文文献人工复核。
10	Ramipril - GLP1R	心血管	机制延展	82.9	有少量文献线索	保留候选, 但购买前需逐篇确认是否已完成直接靶点验证或只是同领域背景共现。
11	Amisulpride - ADRA1A	神经/精神	机制延展	81.3	未见明显直接组合文献	可作为新用途/新靶点候选推进; 购买前仍需 PubMed、ClinicalTrials 和专利/中文文献人工复核。
12	Solifenacin Succinate - CHRM3	神经/精神	阳性对照/已知机制	79.8	已知机制/阳性对照	不作为新机制发现宣传; 建议保留为实验体系校准、阳性对照和召回能力证明。

4 逐项候选评审

4.1 Rank 1: Gefitinib – KIT

候选身份	Gefitinib / KIT / P10721, Mast/stem cell growth factor receptor Kit
疾病方向与角色	肿瘤；机制延展；final gate: go_core_6_after_manual_confirmation；综合分 96.7
已知性标签	本地 knownDrugTargetPair = 未标记为已知；noveltyClass = known_disease_use_with_predicted_target；reviewTrack = A_repurposing_mechanism_review
在线排重信号	PubMed drug-target 共现 131 篇；drug-target-disease 共现 129 篇；ClinicalTrials 药物-疾病方向 471 项
排重结论	已有较多公开共现线索；不建议作为纯新发现表述；应标注为机制延展/二线验证。只有在实验能排除 EGFR carryover 并证明 KIT engagement 时才值得推进。

推荐处理。 Run first if manual checks pass; use this set to calibrate assay behavior and test discovery value. 但需标注为已有较多公开线索的机制延展候选；如果专家更看重全新性，可后移或由 backup 替换。

入选理由。 亲和/模型支持 0.769；疾病证据 OT 1.00, TxGNN 0.00；Reactome/GO/CREEDS 支持靶点疾病上下文；CMap A_strong_reversal, 最佳细胞 MCF10A；GTEx/HPA 相关组织表达支持；DepMap 癌症依赖性支持；结构/口袋审计可解释

证据拆解。 亲和分 0.769；Open Targets 1.000；TxGNN 0.002；KG 证据分 80.0；CMap/CREEDS A_strong_reversal, raw 1.309；通路 A_pathway_signature_supported；组织/模型 A_high_relevant_gtex_expression/A_high_relevant_tissue_expression；DepMap B_subset_cancer_dependency；结构 A_high_confidence_pocket, pose warning, DiffDock -4.15。

机制可解释性。 Type III receptor tyrosine kinase; test KIT biochemical activity and ligand/pathway-dependent cellular signaling. inhibition should suppress KIT signaling in a KIT-expressing model.

实验可行性。 首选实验为 biochemical_kinase_or_enzyme_assay。Primary readout: KIT kinase activity or binding assay, followed by cellular pKIT/pERK/pAKT suppression in KIT-expressing cells. 正交验证: Cell target-engagement readout such as CETSA/NanoBRET or pathway phosphomarker. Counter-screen: Kinase selectivity panel, KIT-low cells, cytotoxicity, and pathway specificity against EGFR/PDGFR where feasible. 推荐模型: KIT-dependent GIST/mast-cell models if available; include KIT-low cells or unrelated RTK cells.

剂量与推进标准。 Start 8-10 point dose response, typically 0.01-30 uM or assay-appropriate range, with vehicle and cytotoxicity controls. Advance only if there is a reproducible dose response at non-toxic concentrations, direct target readout is positive, orthogonal target engagement agrees, and disease phenotype changes in the predicted direction.

停止/降级条件。 Hold if activity appears only with cytotoxicity, no orthogonal target engagement,

poor solubility/exposure, or a counterscreen explains the signal.

主要风险。 Check direct KIT engagement and kinase-family selectivity; avoid interpreting EGFR carryover as KIT biology. ADMET tier A; 模型风险标记: high_herg; moderate_dili; moderate_cyp3a4_veith; moderate_cyp1a2_veith. Gefitinib may show kinase-family cross-reactivity; distinguish direct KIT engagement from EGFR-pathway carryover.

购买前必须确认。 high-priority manual checks: clinical_free_exposure, target_expression_in_model, direction_of_modulation; modulation note: inhibition should suppress KIT signaling in a KIT-expressing model.; assay risk: Check direct KIT engagement and kinase-family selectivity; avoid interpreting EGFR carryover as KIT biology.

专家讨论问题。 Does Gefitinib produce target-specific KIT engagement and a disease-relevant phenotype that cannot be explained by toxicity, assay interference, or the drug's known canonical mechanism?

4.2 Rank 2: Fludarabine Phosphate – ADORA1

候选身份	Fludarabine Phosphate / ADORA1 / P30542, Adenosine receptor A1
疾病方向与角色	肿瘤; 新用途或新靶点候选; final gate: go_core_6_after_manual_confirmation; 综合分 94.8
已知性标签	本地 knownDrugTargetPair = 未标记为已知; noveltyClass = disease_context_supported_new_pair; reviewTrack = B_novel_pair_disease_context_review
在线排重信号	PubMed drug-target 共现 0 篇; drug-target-disease 共现 0 篇; ClinicalTrials 药物-疾病方向 2,465 项
排重结论	未见明显直接组合文献; 可作为新用途/新靶点候选推进; 购买前仍需 PubMed、ClinicalTrials 和专利/中文文献人工复核。

推荐处理。 Run first if manual checks pass; use this set to calibrate assay behavior and test discovery value. 新颖性较高; 优先要求正交靶点 engagement 和 counterscreen 支撑。

入选理由。 亲和/模型支持 0.812; 疾病证据 OT 0.15, TxGNN 0.97; Reactome/GO/CREEDS 支持靶点疾病上下文; CMap B_moderate_reversal, 最佳细胞 HCC515; GTEx/HPA 相关组织表达支持; DepMap 癌症依赖性支持; 结构/口袋审计可解释

证据拆解。 亲和分 0.812; Open Targets 0.146; TxGNN 0.969; KG 证据分 65.0; CMap/CREEDS B_moderate_reversal, raw 0.621; 通路 B_pathway_annotated_context; 组织/模型 A_high_relevant_gtex_expression/A_high_relevant_tissue_expression; DepMap B_subset_cancer_dependency; 结构 A_high_confidence_pocket, pose pass, DiffDock -0.47。

机制可解释性。 Gi-coupled adenosine receptor; validate receptor signaling before interpreting disease-cell phenotype. clarify whether the drug behaves as agonist, antagonist, or indirect modulator in the ADORA1 assay.

实验可行性。 首选实验为 cell_based_receptor_function_assay。Primary readout: ADORA1 functional assay such as inhibition of forskolin-stimulated cAMP, beta-arrestin, or ligand-

competition binding. 正交验证: Ligand competition, receptor internalization, or target knockdown/antagonist rescue. Counter-screen: ADORA2A/ADORA2B/ADORA3 selectivity, receptor-negative cells, nucleoside-transporter context, and cytotoxicity. 推荐模型: ADORA1-overexpressing or endogenous ADORA1-positive cancer/immune/neuronal cells; include receptor-negative parental cells.

剂量与推进标准。 Start 8-10 point dose response, typically 0.01-30 uM or assay-appropriate range, with vehicle and cytotoxicity controls. Advance only if there is a reproducible dose response at non-toxic concentrations, direct target readout is positive, orthogonal target engagement agrees, and disease phenotype changes in the predicted direction.

停止/降级条件。 Hold if activity appears only with cytotoxicity, no orthogonal target engagement, poor solubility/exposure, or a counterscreen explains the signal.

主要风险。 Separate receptor signaling from nucleoside-metabolism or cytotoxic effects. ADMET tier A; 模型风险标记: moderate_dili. Purine analogs can cause nucleoside-metabolism or cytotoxic effects that mimic receptor-linked biology.

购买前必须确认。 high-priority manual checks: clinical_free_exposure, target_expression_in_model, direction_of_modulation; modulation note: clarify whether the drug behaves as agonist, antagonist, or indirect modulator in the ADORA1 assay.; assay risk: Separate receptor signaling from nucleoside-metabolism or cytotoxic effects.

专家讨论问题。 Does Fludarabine Phosphate produce target-specific ADORA1 engagement and a disease-relevant phenotype that cannot be explained by toxicity, assay interference, or the drug's known canonical mechanism?

4.3 Rank 3: Gefitinib – EGFR

候选身份	Gefitinib / EGFR / P00533, Epidermal growth factor receptor
疾病方向与角色	肿瘤; 阳性对照/已知机制; final gate: go_core_6_after_manual_confirmation; 综合分 93.5
已知性标签	本地 knownDrugTargetPair = 已知; noveltyClass = known_mechanism_or_known_disease_use; reviewTrack = A_positive_control_or_known_mechanism
在线排重信号	PubMed drug-target 共现 6,919 篇; drug-target-disease 共现 6,705 篇; ClinicalTrials 药物-疾病方向 471 项
排重结论	已知机制/阳性对照; 不作为新机制发现宣传; 建议保留为实验体系校准、阳性对照和召回能力证明。

推荐处理。 Run first if manual checks pass; use this set to calibrate assay behavior and test discovery value. 该候选属于 阳性对照/已知机制, 在报告中应明确为体系校准而非新发现。发现价值低, 但实验价值高; 若经费紧张, 可以只保留一个同类阳性对照。

入选理由。 亲和/模型支持 0.971; 疾病证据 OT 0.99, TxGNN 0.00; Reactome/GO/CREEDS 支持靶点疾病上下文; CMap A_strong_reversal, 最佳细胞 MCF10A; GTEx/HPA 相关组织表达支持; DepMap

癌症依赖性支持; 结构/口袋审计可解释

证据拆解。 亲和分 0.971; Open Targets 0.991; TxGNN 0.002; KG 证据分 100.0; CMap/CREEDS A_strong_reversal, raw 1.309; 通路 A_pathway_signature_supported; 组织/模型 A_high_relevant_gtex_expression/A_high_relevant_tissue_expression; DepMap A_recurrent_cancer_dependency; 结构 A_high_confidence_pocket, pose pass, DiffDock -0.96。

机制可解释性。 Receptor tyrosine kinase; first validate direct kinase inhibition, then cellular EGFR pathway suppression. inhibition should suppress EGFR pathway signaling; validate pEGFR/pERK or pAKT.

实验可行性。 首选实验为 biochemical_kinase_or_enzyme_assay。Primary readout: EGFR kinase IC50/Kd or enzymatic activity inhibition; pair with pEGFR and downstream pERK/pAKT in EGFR-expressing cells. 正交验证: Cell target-engagement readout such as CETSA/NanoBRET or pathway phospho-marker. Counter-screen: Parallel cytotoxicity, unrelated kinase panel, EGFR-low cells, and pathway-rescue or ligand-stimulation control. 推荐模型: EGFR-driven NSCLC or epithelial cancer lines such as PC9/HCC827 if available; include EGFR-low or EGFR-wild-type cells as specificity context.

剂量与推进标准。 Start 8-10 point dose response, typically 0.01-30 uM or assay-appropriate range, with vehicle and cytotoxicity controls. Advance only if there is a reproducible dose response at non-toxic concentrations, direct target readout is positive, orthogonal target engagement agrees, and disease phenotype changes in the predicted direction.

停止/降级条件。 Hold if activity appears only with cytotoxicity, no orthogonal target engagement, poor solubility/exposure, or a counterscreen explains the signal.

主要风险。 Use EGFR-positive and EGFR-low cells; confirm kinase inhibition rather than broad cytotoxicity. ADMET tier A; 模型风险标记: high_herg; moderate_dili; moderate_cyp3a4_veith; moderate_cyp1a2_veith. Apparent phenotype can be driven by broad kinase/cytotoxic effects rather than target-specific EGFR engagement.

购买前必须确认。 high-priority manual checks: clinical_free_exposure, target_expression_in_model, direction_of_modulation; modulation note: inhibition should suppress EGFR pathway signaling; validate pEGFR/pERK or pAKT.; assay risk: Use EGFR-positive and EGFR-low cells; confirm kinase inhibition rather than broad cytotoxicity.

专家讨论问题。 Does Gefitinib produce target-specific EGFR engagement and a disease-relevant phenotype that cannot be explained by toxicity, assay interference, or the drug's known canonical mechanism?

4.4 Rank 4: Dacomitinib – EGFR

候选身份	Dacomitinib / EGFR / P00533, Epidermal growth factor receptor
疾病方向与角色	肿瘤; 阳性对照/已知机制; final gate: go_core_6_after_manual_confirmation; 综合分 93.2
已知性标签	本地 knownDrugTargetPair = 已知; noveltyClass = known_drug_target_mechanism; reviewTrack = A_positive_control_or_known_mechanism
在线排重信号	PubMed drug-target 共现 374 篇; drug-target-disease 共现 365 篇; ClinicalTrials 药物-疾病方向 62 项
排重结论	已知机制/阳性对照; 不作为新机制发现宣传; 建议保留为实验体系校准、阳性对照和召回能力证明。

推荐处理。 Run first if manual checks pass; use this set to calibrate assay behavior and test discovery value. 该候选属于 阳性对照/已知机制, 在报告中应明确为体系校准而非新发现。发现价值低, 但实验价值高; 若经费紧张, 可以只保留一个同类阳性对照。

入选理由。 亲和/模型支持 0.922; 疾病证据 OT 0.99, TxGNN 0.82; Reactome/GO/CREEDS 支持靶点疾病上下文; CMap A_strong_reversal, 最佳细胞 MCF10A; GTEx/HPA 相关组织表达支持; DepMap 癌症依赖性支持; 结构/口袋审计可解释

证据拆解。 亲和分 0.922; Open Targets 0.991; TxGNN 0.821; KG 证据分 95.0; CMap/CREEDS A_strong_reversal, raw 1.269; 通路 A_pathway_signature_supported; 组织/模型 A_high_relevant_gtex_expression/A_high_relevant_tissue_expression; DepMap A_recurrent_cancer_dependency; 结构 A_high_confidence_pocket, pose pass, DiffDock -0.79。

机制可解释性。 Receptor tyrosine kinase; first validate direct kinase inhibition, then cellular EGFR pathway suppression. inhibition should suppress EGFR pathway signaling; validate pEGFR/pERK or pAKT.

实验可行性。 首选实验为 biochemical_kinase_or_enzyme_assay。Primary readout: EGFR kinase IC50/Kd or enzymatic activity inhibition; pair with pEGFR and downstream pERK/pAKT in EGFR-expressing cells. 正交验证: Cell target-engagement readout such as CETSA/NanoBRET or pathway phospho-marker. Counter-screen: Parallel cytotoxicity, unrelated kinase panel, EGFR-low cells, and pathway-rescue or ligand-stimulation control. 推荐模型: EGFR-driven NSCLC or epithelial cancer lines such as PC9/HCC827 if available; include EGFR-low or EGFR-wild-type cells as specificity context.

剂量与推进标准。 Start 8-10 point dose response, typically 0.01-30 uM or assay-appropriate range, with vehicle and cytotoxicity controls. Advance only if there is a reproducible dose response at non-toxic concentrations, direct target readout is positive, orthogonal target engagement agrees, and disease phenotype changes in the predicted direction.

停止/降级条件。 Hold if activity appears only with cytotoxicity, no orthogonal target engagement, poor solubility/exposure, or a counterscreen explains the signal.

主要风险。 Use EGFR-positive and EGFR-low cells; confirm kinase inhibition rather than broad

cytotoxicity. ADMET tier A; 模型风险标记: moderate_herg; high_dili; moderate_cyp3a4_veith; moderate_cyp2c19_veith; moderate_cyp1a2_veith. Apparent phenotype can be driven by broad kinase/cytotoxic effects rather than target-specific EGFR engagement.

购买前必须确认。 high-priority manual checks: clinical_free_exposure, target_expression_in_model, direction_of_modulation; modulation note: inhibition should suppress EGFR pathway signaling; validate pEGFR/pERK or pAKT.; assay risk: Use EGFR-positive and EGFR-low cells; confirm kinase inhibition rather than broad cytotoxicity.

专家讨论问题。 Does Dacomitinib produce target-specific EGFR engagement and a disease-relevant phenotype that cannot be explained by toxicity, assay interference, or the drug's known canonical mechanism?

4.5 Rank 5: Cladribine – ADORA1

候选身份	Cladribine / ADORA1 / P30542, Adenosine receptor A1
疾病方向与角色	肿瘤; 机制延展; final gate: go_core_6_after_manual_confirmation; 综合分 92.7
已知性标签	本地 knownDrugTargetPair = 未标记为已知; noveltyClass = known_disease_use_with_predicted_target; reviewTrack = A_repurposing_mechanism_review
在线排重信号	PubMed drug-target 共现 1 篇; drug-target-disease 共现 0 篇; ClinicalTrials 药物-疾病方向 147 项
排重结论	有少量文献线索; 保留候选, 但购买前需逐篇确认是否已完成直接靶点验证或只是同领域背景共现。

推荐处理。 Run first if manual checks pass; use this set to calibrate assay behavior and test discovery value. 中等新颖性; 适合与更干净的新组合并行验证。

入选理由。 亲和/模型支持 0.749; 疾病证据 OT 0.15, TxGNN 0.05; Reactome/GO/CREEDS 支持靶点疾病上下文; CMap A_strong_reversal, 最佳细胞 A375; GTEx/HPA 相关组织表达支持; DepMap 癌症依赖性支持; 结构/口袋审计可解释

证据拆解。 亲和分 0.749; Open Targets 0.146; TxGNN 0.046; KG 证据分 80.0; CMap/CREEDS A_strong_reversal, raw 1.172; 通路 B_pathway_annotated_context; 组织/模型 A_high_relevant_gtex_expression/A_high_relevant_tissue_expression; DepMap B_subset_cancer_dependency; 结构 A_high_confidence_pocket, pose pass, DiffDock -0.08.

机制可解释性。 Gi-coupled adenosine receptor; validate receptor signaling before interpreting disease-cell phenotype. clarify whether the drug behaves as agonist, antagonist, or indirect modulator in the ADORA1 assay.

实验可行性。 首选实验为 cell_based_receptor_function_assay。Primary readout: ADORA1 functional assay such as inhibition of forskolin-stimulated cAMP, beta-arrestin, or ligand-competition binding. 正交验证: Ligand competition, receptor internalization, or target knockdown/antagonist rescue. Counter-screen: ADORA2A/ADORA2B/ADORA3 selectivity,

receptor-negative cells, nucleoside-transporter context, and cytotoxicity. 推荐模型: ADORA1-overexpressing or endogenous ADORA1-positive cancer/immune/neuronal cells; include receptor-negative parental cells.

剂量与推进标准。 Start 8-10 point dose response, typically 0.01-30 uM or assay-appropriate range, with vehicle and cytotoxicity controls. Advance only if there is a reproducible dose response at non-toxic concentrations, direct target readout is positive, orthogonal target engagement agrees, and disease phenotype changes in the predicted direction.

停止/降级条件。 Hold if activity appears only with cytotoxicity, no orthogonal target engagement, poor solubility/exposure, or a counterscreen explains the signal.

主要风险。 Separate receptor signaling from nucleoside-metabolism or cytotoxic effects. ADMET tier A; 模型风险标记: high_dili. Purine analogs can cause nucleoside-metabolism or cytotoxic effects that mimic receptor-linked biology.

购买前必须确认。 high-priority manual checks: clinical_free_exposure, target_expression_in_model, direction_of_modulation; modulation note: clarify whether the drug behaves as agonist, antagonist, or indirect modulator in the ADORA1 assay.; assay risk: Separate receptor signaling from nucleoside-metabolism or cytotoxic effects.

专家讨论问题。 Does Cladribine produce target-specific ADORA1 engagement and a disease-relevant phenotype that cannot be explained by toxicity, assay interference, or the drug's known canonical mechanism?

4.6 Rank 6: Palonosetron Hydrochloride – F2R

候选身份	Palonosetron Hydrochloride / F2R / P25116, Proteinase-activated receptor 1
疾病方向与角色	肿瘤; 新用途或新靶点候选; final gate: go_core_6_after_manual_confirmation; 综合分 87.1
已知性标签	本地 knownDrugTargetPair = 未标记为已知; noveltyClass = disease_context_supported_new_pair; reviewTrack = B_novel_pair_disease_context_review
在线排重信号	PubMed drug-target 共现 0 篇; drug-target-disease 共现 0 篇; ClinicalTrials 药物-疾病方向 103 项
排重结论	未见明显直接组合文献; 可作为新用途/新靶点候选推进; 购买前仍需 PubMed、ClinicalTrials 和专利/中文文献人工复核。

推荐处理。 Run first if manual checks pass; use this set to calibrate assay behavior and test discovery value. 新颖性较高; 优先要求正交靶点 engagement 和 counterscreen 支撑。

入选理由。 亲和/模型支持 0.912; 疾病证据 OT 0.00, TxGNN 0.33; Reactome/GO/CREEDS 支持靶点疾病上下文; CMap B_moderate_reversal, 最佳细胞 MCF7; GTEx/HPA 相关组织表达支持; DepMap 癌症依赖性支持; 结构/口袋审计可解释

证据拆解。 亲和分 0.912; Open Targets 0.000; TxGNN 0.331; KG 证据分 50.0; CMap/CREEDS

B_moderate_reversal, raw 0.717; 通路 A_pathway_signature_supported; 组织/模型 A_high_relevant_gtex_expression/A_high_relevant_tissue_expression; DepMap B_subset_cancer_dependency; 结构 A_high_confidence_pocket, pose pass, DiffDock -1.08。

机制可解释性。 PAR1/thrombin receptor GPCR; test receptor-function modulation and separate it from serotonin-receptor pharmacology. clarify PAR1 agonist/antagonist behavior using thrombin or PAR1 activating peptide stimulation.

实验可行性。 首选实验为 cell_based_receptor_function_assay。Primary readout: PAR1 calcium flux, IP1, or beta-arrestin response after thrombin or PAR1 activating peptide stimulation. 正交验证: Ligand competition, receptor internalization, or target knockdown/antagonist rescue. Counter-screen: 5-HT3 receptor counterscreen, cytotoxicity, receptor-negative cells, and thrombin-independent pathway checks. 推荐模型: PAR1-positive tumor, endothelial, platelet-like, or engineered receptor cells; include receptor-low parental cells.

剂量与推进标准。 Start 8-10 point dose response, typically 0.01-30 uM or assay-appropriate range, with vehicle and cytotoxicity controls. Advance only if there is a reproducible dose response at non-toxic concentrations, direct target readout is positive, orthogonal target engagement agrees, and disease phenotype changes in the predicted direction.

停止/降级条件。 Hold if activity appears only with cytotoxicity, no orthogonal target engagement, poor solubility/exposure, or a counterscreen explains the signal.

主要风险。 Counterscreen 5-HT3 biology for palonosetron-like compounds and require receptor-specific rescue/competition. ADMET tier A; 模型风险标记: none。Palonosetron is an established 5-HT3 agent; any PAR1 signal needs strong receptor-specific orthogonal confirmation.

购买前必须确认。 high-priority manual checks: clinical_free_exposure, target_expression_in_model, direction_of_modulation; modulation note: clarify PAR1 agonist/antagonist behavior using thrombin or PAR1 activating peptide stimulation.; assay risk: Counterscreen 5-HT3 biology for palonosetron-like compounds and require receptor-specific rescue/competition.

专家讨论问题。 Does Palonosetron Hydrochloride produce target-specific F2R engagement and a disease-relevant phenotype that cannot be explained by toxicity, assay interference, or the drug's known canonical mechanism?

4.7 Rank 7: Methylnaltrexone Bromide – OXTR

候选身份	Methylnaltrexone Bromide / OXTR / P30559, Oxytocin receptor
疾病方向与角色	神经/精神; 新用途或新靶点候选; final gate: go_extension_12_after_manual_confirmation; 综合分 85.8
已知性标签	本地 knownDrugTargetPair = 未标记为已知; noveltyClass = disease_context_supported_new_pair; reviewTrack = B_novel_pair_disease_context_review
在线排重信号	PubMed drug-target 共现 0 篇; drug-target-disease 共现 0 篇; ClinicalTrials 药物-疾病方向 2 项
排重结论	未见明显直接组合文献; 可作为新用途/新靶点候选推进; 购买前仍需 PubMed、ClinicalTrials 和专利/中文文献人工复核。

推荐处理。 Run after core 6 setup is confirmed, or include immediately if platform capacity is available. 新颖性较高; 优先要求正交靶点 engagement 和 counterscreen 支撑。

入选理由。 亲和/模型支持 0.810; Reactome/GO/CREEDS 支持靶点疾病上下文; CMap B_moderate_reversal, 最佳细胞 PC3; GTEx/HPA 相关组织表达支持; 结构/口袋审计可解释; ADMET A, 安全性可优先审阅

证据拆解。 亲和分 0.810; Open Targets 0.000; TxGNN 0.000; KG 证据分 45.0; CMap/CREEDS B_moderate_reversal, raw 0.704; 通路 A_pathway_signature_supported; 组织/模型 B_moderate_relevant_gtex_expression/A_high_relevant_tissue_expression; DepMap C_weak_or_rare_dependency; 结构 A_high_confidence_pocket, pose pass, DiffDock -1.90。

机制可解释性。 Gq-coupled oxytocin receptor; validate receptor-function modulation and separate from opioid/peripheral effects. clarify OXTR agonist/antagonist behavior through calcium/IP1/beta-arrestin response.

实验可行性。 首选实验为 cell_based_receptor_function_assay。Primary readout: OXTR calcium flux, IP1, or beta-arrestin response to oxytocin with agonist/antagonist dose-response. 正交验证: Ligand competition, receptor internalization, or target knockdown/antagonist rescue. Counter-screen: AVPR1A/AVPR1B/AVPR2 selectivity, opioid receptor counterscreen, cytotoxicity, and receptor-negative cells. 推荐模型: OXTR-expressing engineered, neuronal, endocrine, or smooth-muscle models after expression confirmation.

剂量与推进标准。 Start 8-10 point dose response, typically 0.01-30 uM or assay-appropriate range, with vehicle and cytotoxicity controls. Advance only if there is a reproducible dose response at non-toxic concentrations, direct target readout is positive, orthogonal target engagement agrees, and disease phenotype changes in the predicted direction.

停止/降级条件。 Hold if activity appears only with cytotoxicity, no orthogonal target engagement, poor solubility/exposure, or a counterscreen explains the signal.

主要风险。 Check permeability and opioid receptor counterscreens for methylnaltrexone-like compounds. ADMET tier A; 模型风险标记: none。Methylnaltrexone is quaternary/peripherally acting; cell permeability and opioid receptor effects must be checked.

购买前必须确认。 high-priority manual checks: clinical_free_exposure, target_expression_in_model, direction_of_modulation; modulation note: clarify OXTR agonist/antagonist behavior through calcium/IP1/beta-arrestin response.; assay risk: Check permeability and opioid receptor counterscreens for methylnaltrexone-like compounds.

专家讨论问题。 Does Methylnaltrexone Bromide produce target-specific OXTR engagement and a disease-relevant phenotype that cannot be explained by toxicity, assay interference, or the drug's known canonical mechanism?

4.8 Rank 8: Methylnaltrexone Bromide – CRHR1

候选身份	Methylnaltrexone Bromide / CRHR1 / P34998, Corticotropin-releasing factor receptor 1
疾病方向与角色	神经/精神; 新用途或新靶点候选; final gate: go_extension_12_after_manual_confirmation; 综合分 85.0
已知性标签	本地 knownDrugTargetPair = 未标记为已知; noveltyClass = disease_context_supported_new_pair; reviewTrack = B_novel_pair_disease_context_review
在线排重信号	PubMed drug-target 共现 0 篇; drug-target-disease 共现 0 篇; ClinicalTrials 药物-疾病方向 2 项
排重结论	未见明显直接组合文献; 可作为新用途/新靶点候选推进; 购买前仍需 PubMed、ClinicalTrials 和专利/中文文献人工复核。

推荐处理。 Run after core 6 setup is confirmed, or include immediately if platform capacity is available. 新颖性较高; 优先要求正交靶点 engagement 和 counterscreen 支撑。

入选理由。 亲和/模型支持 0.783; Reactome/GO/CREEDS 支持靶点疾病上下文; CMap B_moderate_reversal, 最佳细胞 PC3; GTEx/HPA 相关组织表达支持; 结构/口袋审计可解释; ADMET A, 安全性可优先审阅

证据拆解。 亲和分 0.783; Open Targets 0.000; TxGNN 0.000; KG 证据分 45.0; CMap/CREEDS B_moderate_reversal, raw 0.704; 通路 A_pathway_signature_supported; 组织/模型 A_high_relevant_gtex_expression/A_high_relevant_tissue_expression; DepMap C_weak_or_rare_dependency; 结构 A_high_confidence_pocket, pose pass, DiffDock -3.94。

机制可解释性。 Gs-coupled corticotropin-releasing hormone receptor; validate cAMP signaling before disease-cell phenotype. antagonist activity should reduce CRH/urocortin-induced CRHR1 cAMP signaling.

实验可行性。 首选实验为 cell_based_receptor_function_assay。Primary readout: CRHR1 cAMP response to CRH/urocortin and antagonist-mode dose-response. 正交验证: Ligand competition, receptor internalization, or target knockdown/antagonist rescue. Counter-screen: CRHR2 selectivity, opioid receptor counterscreen, cytotoxicity, and receptor-negative cells. 推荐模型: CRHR1-expressing engineered, neuronal, endocrine, or stress-axis relevant cells with target expression confirmed.

剂量与推进标准。 Start 8-10 point dose response, typically 0.01-30 uM or assay-appropriate range, with vehicle and cytotoxicity controls. Advance only if there is a reproducible dose response at non-toxic concentrations, direct target readout is positive, orthogonal target engagement agrees, and disease phenotype changes in the predicted direction.

停止/降级条件。 Hold if activity appears only with cytotoxicity, no orthogonal target engagement, poor solubility/exposure, or a counterscreen explains the signal.

主要风险。 Counterscreen CRHR2 and canonical opioid pharmacology where relevant. ADMET tier A; 模型风险标记: none。Methylnaltrexone' s canonical opioid pharmacology and permeability can create false receptor-phenotype links.

购买前必须确认。 high-priority manual checks: clinical_free_exposure, target_expression_in_model, direction_of_modulation; modulation note: antagonist activity should reduce CRH/urocortin-induced CRHR1 cAMP signaling.; assay risk: Counterscreen CRHR2 and canonical opioid pharmacology where relevant.

专家讨论问题。 Does Methylnaltrexone Bromide produce target-specific CRHR1 engagement and a disease-relevant phenotype that cannot be explained by toxicity, assay interference, or the drug' s known canonical mechanism?

4.9 Rank 9: Donepezil Hydrochloride – EDNRA

候选身份	Donepezil Hydrochloride / EDNRA / P25101, Endothelin-1 receptor
疾病方向与角色	心血管; 新用途或新靶点候选; final gate: go_extension_12_after_manual_confirmation; 综合分 84.7
已知性标签	本地 knownDrugTargetPair = 未标记为已知; noveltyClass = disease_context_supported_new_pair; reviewTrack = B_novel_pair_disease_context_review
在线排重信号	PubMed drug-target 共现 0 篇; drug-target-disease 共现 0 篇; ClinicalTrials 药物-疾病方向 63 项
排重结论	未见明显直接组合文献; 可作为新用途/新靶点候选推进; 购买前仍需 PubMed、ClinicalTrials 和专利/中文文献人工复核。

推荐处理。 Run after core 6 setup is confirmed, or include immediately if platform capacity is available. 新颖性较高; 优先要求正交靶点 engagement 和 counterscreen 支撑。

入选理由。 亲和/模型支持 0.902; Reactome/GO/CREEDS 支持靶点疾病上下文; CMap B_moderate_reversal, 最佳细胞 JURKAT; GTEx/HPA 相关组织表达支持; 结构/口袋审计可解释; ADMET A, 安全性可优先审阅

证据拆解。 亲和分 0.902; Open Targets 0.000; TxGNN 0.000; KG 证据分 40.0; CMap/CREEDS B_moderate_reversal, raw 0.866; 通路 A_pathway_signature_supported; 组织/模型 A_high_relevant_gtex_expression/A_high_relevant_tissue_expression; DepMap B_subset_cancer_dependency; 结构 B_moderate_confidence_pocket, pose pass, DiffDock -2.23。

机制可解释性。 Endothelin A receptor GPCR; useful for cardiovascular/vascular phenotype only

after direct receptor readout. antagonist activity should reduce endothelin-1-induced EDNRA signaling.

实验可行性。 首选实验为 cell_based_receptor_function_assay。Primary readout: EDNRA calcium flux, IP1, or beta-arrestin response to endothelin-1 and antagonist-mode dose-response. 正交验证: Ligand competition, receptor internalization, or target knockdown/antagonist rescue. Counter-screen: EDNRB selectivity, cholinesterase/pathway counterscreen for donepezil, cytotoxicity, and receptor-negative cells. 推荐模型: EDNRA-positive vascular smooth-muscle, endothelial, cardiomyocyte, or engineered receptor cells.

剂量与推进标准。 Start 8-10 point dose response, typically 0.01-30 uM or assay-appropriate range, with vehicle and cytotoxicity controls. Advance only if there is a reproducible dose response at non-toxic concentrations, direct target readout is positive, orthogonal target engagement agrees, and disease phenotype changes in the predicted direction.

停止/降级条件。 Hold if activity appears only with cytotoxicity, no orthogonal target engagement, poor solubility/exposure, or a counterscreen explains the signal.

主要风险。 Separate EDNRA engagement from cholinesterase-related pharmacology for donepezil-like compounds. ADMET tier A; 模型风险标记: high_herg; moderate_cyp3a4_veith; moderate_cyp2d6_veith; moderate_pgp_broccatelli. Donepezil's canonical cholinesterase activity can confound cardiovascular phenotypes without direct EDNRA engagement.

购买前必须确认。 high-priority manual checks: clinical_free_exposure, target_expression_in_model, direction_of_modulation; modulation note: antagonist activity should reduce endothelin-1-induced EDNRA signaling.; assay risk: Separate EDNRA engagement from cholinesterase-related pharmacology for donepezil-like compounds.

专家讨论问题。 Does Donepezil Hydrochloride produce target-specific EDNRA engagement and a disease-relevant phenotype that cannot be explained by toxicity, assay interference, or the drug's known canonical mechanism?

4.10 Rank 10: Ramipril – GLP1R

候选身份	Ramipril / GLP1R / P43220, Glucagon-like peptide 1 receptor
疾病方向与角色	心血管; 机制延展; final gate: go_extension_12_after_manual_confirmation; 综合分 82.9
已知性标签	本地 knownDrugTargetPair = 未标记为已知; noveltyClass = known_disease_use_with_predicted_target; reviewTrack = A_repurposing_mechanism_review
在线排重信号	PubMed drug-target 共现 3 篇; drug-target-disease 共现 1 篇; ClinicalTrials 药物-疾病方向 167 项
排重结论	有少量文献线索; 保留候选, 但购买前需逐篇确认是否已完成直接靶点验证或只是同领域背景共现。

推荐处理。 Run after core 6 setup is confirmed, or include immediately if platform capacity is

available. 中等新颖性；适合与更干净的新组合并行验证。

入选理由。 亲和/模型支持 0.682；Reactome/GO/CREEDS 支持靶点疾病上下文；CMap B_moderate_reversal, 最佳细胞 A375；GTEx/HPA 相关组织表达支持；结构/口袋审计可解释；ADMET A, 安全性可优先审阅

证据拆解。 亲和分 0.681；Open Targets 0.000；TxGNN 0.000；KG 证据分 40.0；CMap/CREEDS B_moderate_reversal, raw 0.829；通路 A_pathway_signature_supported；组织/模型 B_moderate_relevant_gtex_expression/B_moderate_relevant_tissue_expression；DepMap D_no_depmap_dependency_signal；结构 A_high_confidence_pocket, pose pass, DiffDock -3.51。

机制可解释性。 Gs-coupled peptide GPCR; receptor engagement should be shown through cAMP or beta-arrestin before disease interpretation. clarify GLP1R agonist/antagonist mode through cAMP or beta-arrestin response.

实验可行性。 首选实验为 cell_based_receptor_function_assay。Primary readout: GLP1R cAMP accumulation or beta-arrestin response with agonist/antagonist mode clarified. 正交验证: Ligand competition, receptor internalization, or target knockdown/antagonist rescue. Counter-screen: ACE-pathway counterscreen for ramipril, GLP1R-negative cells, cytotoxicity, and peptide-receptor specificity. 推荐模型: GLP1R-expressing engineered cells, endocrine/metabolic cells, endothelial/cardiometabolic models if expression is confirmed.

剂量与推进标准。 Start 8-10 point dose response, typically 0.01-30 uM or assay-appropriate range, with vehicle and cytotoxicity controls. Advance only if there is a reproducible dose response at non-toxic concentrations, direct target readout is positive, orthogonal target engagement agrees, and disease phenotype changes in the predicted direction.

停止/降级条件。 Hold if activity appears only with cytotoxicity, no orthogonal target engagement, poor solubility/exposure, or a counterscreen explains the signal.

主要风险。 For ramipril-like hypotheses, separate receptor signal from ACE-pathway and metabolite effects. ADMET tier A；模型风险标记: moderate_herg; moderate_dili; moderate_pgp_broccatelli. Ramipril is a prodrug/ACE-pathway drug; receptor signal must be separated from canonical ACE biology and metabolite effects.

购买前必须确认。 high-priority manual checks: clinical_free_exposure, target_expression_in_model, direction_of_modulation; modulation note: clarify GLP1R agonist/antagonist mode through cAMP or beta-arrestin response.; assay risk: For ramipril-like hypotheses, separate receptor signal from ACE-pathway and metabolite effects.

专家讨论问题。 Does Ramipril produce target-specific GLP1R engagement and a disease-relevant phenotype that cannot be explained by toxicity, assay interference, or the drug's known canonical mechanism?

4.11 Rank 11: Amisulpride – ADRA1A

候选身份	Amisulpride / ADRA1A / P35348, Alpha-1A adrenergic receptor
疾病方向与角色	神经/精神; 机制延展; final gate: go_extension_12_after_manual_confirmation; 综合分 81.3
已知性标签	本地 knownDrugTargetPair = 未标记为已知; noveltyClass = known_disease_use_with_predicted_target; reviewTrack = A_repurposing_mechanism_review
在线排重信号	PubMed drug-target 共现 0 篇; drug-target-disease 共现 0 篇; ClinicalTrials 药物-疾病方向 51 项
排重结论	未见明显直接组合文献; 可作为新用途/新靶点候选推进; 购买前仍需 PubMed、ClinicalTrials 和专利/中文文献人工复核。

推荐处理。 Run after core 6 setup is confirmed, or include immediately if platform capacity is available. 新颖性较高; 优先要求正交靶点 engagement 和 counterscreen 支撑。

入选理由。 亲和/模型支持 0.834; Reactome/GO/CREEDS 支持靶点疾病上下文; CMap C_weak_or_context_dependent_reversal, 最佳细胞 MCLF135CN; GTEx/HPA 相关组织表达支持; 结构/口袋审计可解释; ADMET A, 安全性可优先审阅

证据拆解。 亲和分 0.834; Open Targets 0.000; TxGNN 0.000; KG 证据分 70.0; CMap/CREEDS C_weak_or_context_dependent_reversal, raw 0.665; 通路 A_pathway_signature_supported; 组织/模型 B_moderate_relevant_gtex_expression/B_moderate_relevant_tissue_expression; DepMap D_no_depmap_dependency_signal; 结构 B_moderate_confidence_pocket, pose pass, DiffDock -2.40。

机制可解释性。 Gq-coupled adrenergic receptor; validate receptor-function modulation and subtype selectivity. antagonist activity should reduce phenylephrine/norepinephrine-induced alpha-1A signaling.

实验可行性。 首选实验为 cell_based_receptor_function_assay。Primary readout: ADRA1A calcium flux or IP1 response to phenylephrine/norepinephrine with antagonist-mode dose-response. 正交验证: Ligand competition, receptor internalization, or target knockdown/antagonist rescue. Counter-screen: ADRA1B/ADRA1D, dopamine receptor, cytotoxicity, and parental-cell counterscreens. 推荐模型: ADRA1A-expressing engineered cells or neuronal/vascular models with confirmed expression.

剂量与推进标准。 Start 8-10 point dose response, typically 0.01-30 uM or assay-appropriate range, with vehicle and cytotoxicity controls. Advance only if there is a reproducible dose response at non-toxic concentrations, direct target readout is positive, orthogonal target engagement agrees, and disease phenotype changes in the predicted direction.

停止/降级条件。 Hold if activity appears only with cytotoxicity, no orthogonal target engagement, poor solubility/exposure, or a counterscreen explains the signal.

主要风险。 Counterscreen dopaminergic pharmacology for amisulpride-like compounds. ADMET tier A; 模型风险标记: moderate_herg. Amisulpride has dopaminergic pharmacology;

alpha-1 signal needs clean receptor-selectivity evidence.

购买前必须确认。 high-priority manual checks: clinical_free_exposure, target_expression_in_model, direction_of_modulation; modulation note: antagonist activity should reduce phenylephrine/norepinephrine-induced alpha-1A signaling.; assay risk: Counterscreen dopaminergic pharmacology for amisulpride-like compounds.

专家讨论问题。 Does Amisulpride produce target-specific ADRA1A engagement and a disease-relevant phenotype that cannot be explained by toxicity, assay interference, or the drug's known canonical mechanism?

4.12 Rank 12: Solifenacin Succinate – CHRM3

候选身份	Solifenacin Succinate / CHRM3 / P20309, Muscarinic acetylcholine receptor M3
疾病方向与角色	神经/精神; 阳性对照/已知机制; final gate: go_extension_12_after_manual_confirmation; 综合分 79.8
已知性标签	本地 knownDrugTargetPair = 已知; noveltyClass = known_drug_target_mechanism; reviewTrack = A_positive_control_or_known_mechanism
在线排重信号	PubMed drug-target 共现 5 篇; drug-target-disease 共现 0 篇; ClinicalTrials 药物-疾病方向 21 项
排重结论	已知机制/阳性对照; 不作为新机制发现宣传; 建议保留为实验体系校准、阳性对照和召回能力证明。

推荐处理。 Run after core 6 setup is confirmed, or include immediately if platform capacity is available. 该候选属于 阳性对照/已知机制, 在报告中应明确为体系校准而非新发现。发现价值低, 但实验价值高; 若经费紧张, 可以只保留一个同类阳性对照。

入选理由。 亲和/模型支持 0.800; Reactome/GO/CREEDS 支持靶点疾病上下文; CMap B_moderate_reversal, 最佳细胞 HA1E; GTEx/HPA 相关组织表达支持; 结构/口袋审计可解释; ADMET A, 安全性可优先审阅

证据拆解。 亲和分 0.800; Open Targets 0.000; TxGNN 0.000; KG 证据分 55.0; CMap/CREEDS B_moderate_reversal, raw 0.924; 通路 A_pathway_signature_supported; 组织/模型 B_moderate_relevant_gtex_expression/B_moderate_relevant_tissue_expression; DepMap D_no_depmap_dependency_signal; 结构 A_high_confidence_pocket, pose pass, DiffDock -2.45。

机制可解释性。 Gq-coupled muscarinic receptor; useful as receptor-function calibration and antimuscarinic assay control. antagonist activity should reduce acetylcholine/carbachol-induced CHRM3 signaling.

实验可行性。 首选实验为 cell_based_receptor_function_assay。Primary readout: CHRM3 calcium flux or IP1 response to acetylcholine/carbachol and antagonist dose-response. 正交验证: Ligand competition, receptor internalization, or target knockdown/antagonist rescue. Counter-screen: Muscarinic subtype selectivity, parental-cell response, cytotoxicity, and assay-

interference controls. 推荐模型: CHRM3-expressing engineered cells or endogenous epithelial/smooth-muscle models; include CHRM1/2/4/5 selectivity if feasible.

剂量与推进标准。 Start 8-10 point dose response, typically 0.01-30 uM or assay-appropriate range, with vehicle and cytotoxicity controls. Advance only if there is a reproducible dose response at non-toxic concentrations, direct target readout is positive, orthogonal target engagement agrees, and disease phenotype changes in the predicted direction.

停止/降级条件。 Hold if activity appears only with cytotoxicity, no orthogonal target engagement, poor solubility/exposure, or a counterscreen explains the signal.

主要风险。 Use muscarinic subtype counterscreens because subtype spillover is common. ADMET tier A; 模型风险标记: moderate_herg. Subtype selectivity can be weak in cell assays; use orthogonal ligand competition or subtype panel.

购买前必须确认。 high-priority manual checks: clinical_free_exposure, target_expression_in_model, direction_of_modulation; modulation note: antagonist activity should reduce acetylcholine/carbachol-induced CHRM3 signaling.; assay risk: Use muscarinic subtype counterscreens because subtype spillover is common.

专家讨论问题。 Does Solifenacin Succinate produce target-specific CHRM3 engagement and a disease-relevant phenotype that cannot be explained by toxicity, assay interference, or the drug’ s known canonical mechanism?

5 替补候选索引

以下候选未写成逐项实验计划，但可在阳性对照过多、已有研究过强、采购不可行或模型表达不合适时替换进入专家审阅。

序号	候选	方向	角色	分数	作为替补的理由
1	Cladribine - GCGR	肿瘤	机制延展	93.0	all automatic evidence gates passed
2	Imatinib Mesylate - KIT	肿瘤	阳性对照/已知机制	88.1	all automatic evidence gates passed
3	Solifenacin Succinate - CHRM2	神经/精神	二线专家复核候选	87.3	all automatic evidence gates passed
4	Dasatinib - FYN	感染性疾病	二线专家复核候选	87.0	all automatic evidence gates passed
5	Dasatinib - YES1	免疫/炎症	二线专家复核候选	86.7	all automatic evidence gates passed
6	Atropine - CHRM1	神经/精神	二线专家复核候选	84.3	failed automatic gates: signature_reversal
7	Bosutinib Monohydrate - FYN	感染性疾病	新用途或新靶点候选	81.8	failed automatic gates: disease_context

8	Atropine - DRD3	神 经/ 精神	新用途或新靶点候选	81.8	failed automatic gates: signature_reversal
9	Palonosetron Hydrochloride - F2R	Hy- 免 疫/ 炎症	新用途或新靶点候选	79.5	failed automatic gates: disease_context
10	Sumatriptan Succinate - SSTR3	Succi- 免 疫/ 炎症	新用途或新靶点候选	77.4	failed automatic gates: disease_context
11	Tivozanib Hydrochloride - CSF1R	Hy- 感 染 性 疾病	新用途或新靶点候选	77.3	failed automatic gates: disease_context

6 统一的 Go/No-Go 原则

- 只有在非毒性浓度下出现可复现剂量反应，且 primary target readout 与正交 target engagement/pathway readout 方向一致时，才进入下一轮。
- 对 receptor 类候选，必须区分 agonist、antagonist 或间接调节；对 kinase/enzyme 类候选，必须加入同家族/相关通路 counterscreen。
- 对已知机制或已有大量文献线索的候选，只能作为阳性对照、体系校准或机制延展，不应包装为新发现。
- 对 CMap/CREEDS 支持但靶点 engagement 不明确的候选，不应仅凭疾病表型推进，必须先证明靶点相关性。
- 对 ADMET 风险标记较高或临床暴露明显不可达的候选，应先做低成本体外验证，不建议直接进入复杂疾病模型。

7 数据源

- 候选与模型证据：final gate Top12 CSV 与完整 gate CSV。
- 同目录产物：文献排重 CSV、Markdown 源和本 PDF。
- 在线检索源：[NCBI PubMed ESearch API](#)；[ClinicalTrials.gov API v2](#)。查询日期为 2026-06-05。